The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 May 2008.

INCH-POUND
MIL-PRF-19500/567D
8 February 2008
SUPERSEDING
MIL-PRF-19500/567C
12 September 2003

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON,
SCHOTTKY BARRIER, FAST RECOVERY,
TYPE 1N6492, 1N6492U4, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, fast recovery, Schottky barrier, semiconductor diode. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

* 1.2 Physical dimensions. See figure 1 (TO-205AF, formerly low profile TO-39), and figure 2 (U4).

* 1.3 Maximum ratings.

Type	V_{RRM} and V_{RWM} (1)	V_{RSM}	V_R (1)	I_F 1 (AV) T_A = +25°C (2)	I_F 1 (AV) T_C = +100°C (3)	I_O T_C = +100°C (4)	I_{FSM}	$R_{\Theta JC}$ (5)	$R_{\Theta JA}$ (5)	T_J and T_{STG}
	V (pk)	V (pk)	V dc	A dc	A dc	A dc	A (pk)	°C/W	°C/W	°C
1N6492	45	54	45	1.20	4	3.60	80	12.0	175	-65 to
1N6492U4	45	54	45	1.20	4	3.60	80	12.0	175	+175

(1) Full rated V_{RRM} and V_{RWM} with appropriate average forward current (see note (3)) is applicable over the range of T_C from -55°C to +135°C. Full rated V_R is applicable over the range of T_C from -55°C to +120°C. With these maximum voltages and case temperatures, $T_J \leq$ +175°C.
(2) This rating requires no special mounting, heat sinking, or forced air flow across the device.
(3) Average current with a 50 percent duty cycle square wave including reverse voltage amplitude equal to the magnitude of full rated V_{RWM}. Derate linearly at 114 mA dc/°C for T_C > +100°C (to 0 at T_C = +135°C); if V_{RWM} = 20, derate I_F (AV) at 62 mA/°C, to 0 at T_C = +165°C.
(4) Average current with an applied sine wave including reverse voltage equal to the magnitude of full rated V_{RWM}. Derate linearly at 103 mA dc/°C for T_C > +100°C; if V_{RWM} = 20, derate at 55 mA/°C.
(5) For thermal impedance see figure 3.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to semiconductor@dscc.dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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* 1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise indicated.

Type	V_{FM1} I_{FM} = 4 A (pk)	V_{FM2} I_{FM} = 2 A (pk)	I_{RM} V_{RM} = 45 V (pk) T_J = +125°C	I_{RM} V_{RM} = 45 V (pk) T_J = +25°C	C_T V_R = 5 V dc
	V (pk)	V (pk)	mA (pk)	mA (pk)	pF
1N6492	.68	.56	20	2.0	450
1N6492U4	.68	.56	20	2.0	450

2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

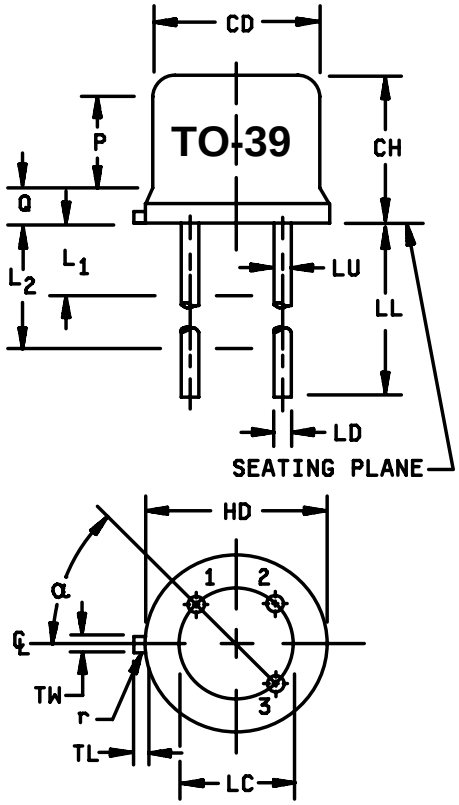
3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

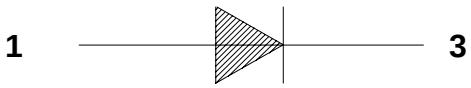
3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 2 herein.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.160	.180	4.07	4.57	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		7
LD	.016	.021	0.41	0.53	8, 9
LL	.500	.750	12.7	19.05	8, 9
LU	.016	.019	0.41	0.48	8, 9
L_1		.050		1.27	8, 9
L_2	.250		6.35		8, 9
P	.100		2.54		6
Q		.040		1.02	5
r		.010		0.254	10
TL	.029	.045	0.74	1.14	
TW	.028	.034	0.72	0.86	
α	45° TP		45° TP		7
Term 1	Anode				
Term 2	Open (no connection)				
Term 3	Cathode (case)				



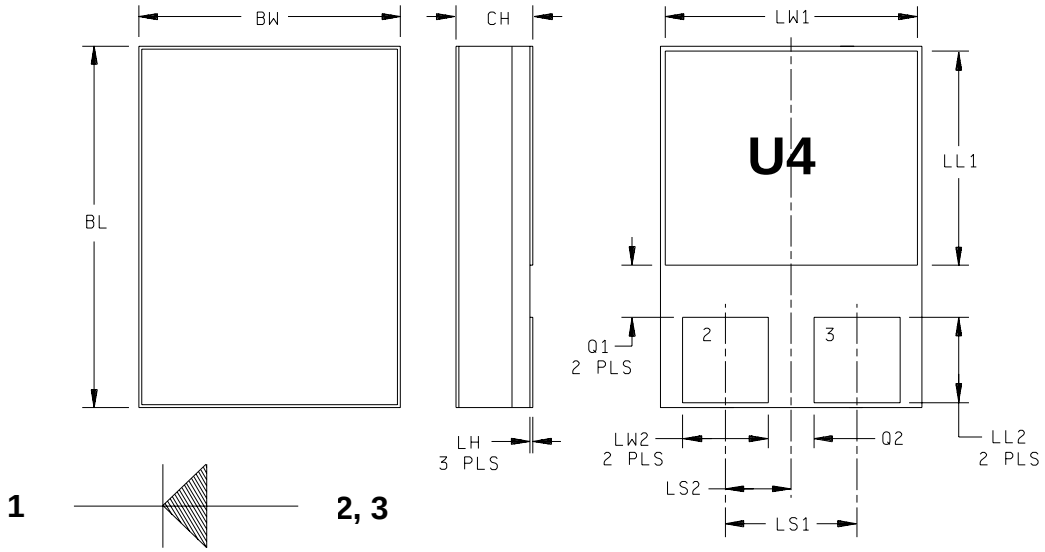


NOTES:

1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Beyond radius (r) maximum, TW shall be held for a minimum length of .011 inch (0.279 mm).
4. Dimension TL measured from maximum HD.
5. Outline in this zone is not controlled.
6. Dimension CD shall not vary more than .010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
7. Leads at gauge plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
8. LU applies between L_1 and L_2. LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
9. All three leads.
10. Radius (r) applies to both inside corners of tab.
11. Cathode is electrically connected to the case.
12. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

* FIGURE 1. Physical dimensions (TO-205AF - formerly low profile TO-39).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	0.215	0.225	5.46	5.72
BW	0.145	0.155	3.68	3.94
CH	0.049	0.075	1.24	1.91
LH	-	0.020	-	0.508
LL1	0.085	0.125	2.16	3.17
LL2	0.045	0.075	1.14	1.90
LS1	0.070	0.095	1.78	2.41
LS2	0.035	0.048	0.889	1.21
LW1	0.135	0.145	3.43	3.68
LW2	0.047	0.057	1.19	1.45
Q1	0.030	0.070	0.762	1.78
Q2	0.020	0.035	0.508	0.88
TERM 1	Cathode			
TERM 2	Anode 1			
TERM 3	Anode 2			

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

* FIGURE 2. Physical dimensions and configuration (U4).

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3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with figure 4 of MIL-PRF-19500.

* 4.3 Screening (JANS, JANTXV, and JANTX levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3	Test condition F, T_{LOW} = -55°C and T_{HIGH} = +175°C.	Test condition F, T_{LOW} = -55°C and T_{HIGH} = +175°C.
9	Reverse energy test (see 4.5.2) V_{FM2} , I_{RM1}.	Not applicable.
10	T_C = +130°C.	T_C = +130°C.
11	Reverse energy test (see 4.5.2), ΔV_{FM2} ±50 mV of initial value, ΔI_{RM1} ±100 percent or 500 μA, whichever is greater.	Reverse energy test (see 4.5.2), V_{FM2} , I_{RM1}.
12	See 4.3.1.	See 4.3.1.
13	Subgroup 2 of table I herein. Reverse energy test (see 4.5.2), ΔV_{FM2} ±50 mV, ΔI_{RM1} ±100 percent or 500 μA, whichever is greater. Scope display evaluation (see 4.5.5).	Subgroup 2 of table I herein. Reverse energy test (see 4.5.2), ΔV_{FM2} ±50 mV, ΔI_{RM1} ±100 percent or 500 μA, whichever is greater. Scope display evaluation (see 4.5.5).

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $V_R = V_{RWM}$. T_A = +50°C maximum, any clips or heat sink configuration may be utilized provided that I_O and T_A are adjusted to achieve T_J = +135°C minimum, V_R = 45 V (pk), I_O = 0.75 A dc minimum, f = 60 Hz. Mounting and test conditions shall be in accordance with Method 1038 of MIL-STD-750 , test condition B.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with figure 4 of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) and VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and 4.4.2.1 and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table E-VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1051	Temperature cycling, condition C, T_{LOW} = -55°C, T_{HIGH} = +175°C.
B4	1037	I_O = 0.75 A, T_A = +25°C; V_{RM} = 45 V, t_{on} = t_{off} = 3 minutes for a minimum of 2,000 cycles.
B5	1027	I_F = 3.2 A dc minimum adjust T_A or I_F as required I_F or T_A to achieve a lot T_J = +275°C.
B6	4081	See 4.5.3, $R_{\Theta JC} \leq$ 12.0°C/W.

4.2.2 Group B inspection, table E-VIb (JAN, JANTX, and JANTXV of MIL-PRF-19500).

Subgroup	Method	Condition
B2	1051	Temperature cycling, condition C.
B3	1027	T_A = +25°C, I_O = 0.75 A, V_{RM} = 45 V (pk).
B3	2037	Condition A, all internal wires must be pulled separately.
B5	4081	See 4.5.3, $R_{\Theta JC} \leq$ 12.0°C/W.
B6	1032	T_A = +175°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

* 4.4.3.1 Group C inspection, table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C6	1026	I_O = 0.75 A, V_{RWN} = 45 V(pk), T_A = +25°C.
* C6		Operational power cycling, see 4.5.6, $T_{C(LOW)}$ = +40°C, +0, -15°C; $T_{C(HIGH)}$ = +115°C +5, -0°C, 5,000 cycles, n = 22, c = 0.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Peak reverse energy test. The peak reverse energy test is to be performed as shown on figure 4 or equivalent. The diode under test must be capable of absorbing the reverse energy, as defined, and meet the electrical requirements of table I, subgroup 4 herein.

4.5.3 Thermal resistance. Thermal resistance shall be measured in accordance with method 4081 of MIL-STD-750. The case reference temperature shall be held to equilibrium within the range of +20°C to +70°C during the power application, and shall be measured at the hottest part of the case. The following measurements shall apply: I_{F1} = 1 A, at 97 percent minimum duty factor; I_{F2} = 10 mA dc.

4.5.4 Reverse current at peak reverse voltage, alternate test. The reverse current at peak reverse voltage test may be satisfied by performing the reverse energy test of 4.5.2 and measuring breakdown voltage to ensure $V_{(BR)} \geq$ 54 V (pk) with I_{RM} = 2.0 A (pk). See figure 4.

4.5.5 Scope display evaluation. Scope display evaluation shall be sharp and stable in accordance with method 4023 of MIL-STD-750. Scope display evaluation may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display evaluation in group A shall be performed on an oscilloscope. One hundred percent scope test is required in the event of a group A failure, however group A resubmission criteria applies.

4.5.6 Operational power cycling (qualification only). One complete cycle for this test shall consist of the following two steps: Step 1, heat the case to the $T_{C(high)}$ specified, by passing forward current through the diode under test. The reverse voltage shall be only enough to permit the reverse current to flow, and should be a maximum of 5 volts. Step 2, remove the applied current and allow the case temperature to cool to the $T_{C(LOW)}$ specified. Forward current shall be chosen to achieve the $T_{c(high)}$ condition in 75 $\pm$50 seconds. The cycling must be continuous until the required number of cycles has been completed. It is permissible to force cool the device during step 2.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2		2/				
Forward voltage	4011	I_{FM} = 4.0 A(pk), pulsed (see 4.5.1)	V_{FM2}		0.68	V
	4011	I_{FM} = 2.0 A(pk), pulsed (see 4.5.1)	V_{FM3}		0.56	V
	4011	I_{FM} = 1.0 A(pk), pulsed (see 4.5.1)	V_{FM4}		0.48	V
	4011	I_{FM} = 8.0 A(pk), pulsed (see 4.5.1)	V_{FM1}		0.92	V
Reverse current leakage	4016	V_{RM} = 45 V(pk), pulsed method, (see 4.5.1)	I_{RM1}		2.0	mA
Subgroup 3		2/				
High temperature operation		T_A = +125°C				
Reverse current leakage	4016	V_{RM}= 45 V(pk), pulsed method, (see 4.5.1)	I_{RM2}		20	mA
High temperature operation		T_A = +175°C				
Reverse current leakage	4016	V_{RM}= 45 V(pk), pulsed method, (see 4.5.1)	I_{RM3}		200	mA
Low temperature operation		T_A = -55°C				
Reverse current leakage	4016	V_{RM} = 45 V(pk), pulsed method, (see 4.5.1)	I_{RM4}		20	mA
Forward voltage	4011	I_{FM} = 2.0 A(pk), pulsed (see 4.5.1)	V_{FM5}		0.63	V
Subgroup 4						
Reverse current leakage at peak reverse voltage	4016	Pulsed method (see 4.5.1), V_{RSM} = 54 V(pk), (alternate test, see 4.5.4) 2/	I_{RM5}		2.0	A
Capacitance	4001	V_R = 5 V, .01 ≤ f ≤ 1 MHz, V_{SIG} = 15 mV(p-p)	C_T		450	pF
Scope display evaluation	4023	Sharp and stable (see 4.5.5), n = 116, c = 0				
Subgroup 5						
Surge current	4066	I_{FSM}= 80 A(pk), V_{RM} = 45 V(pk), I_O = 0.75 A ten surges of 8.3 ms each at 1 minute intervals, T_A = +25°C				
Electrical measurements		See table I, subgroup 2 herein				

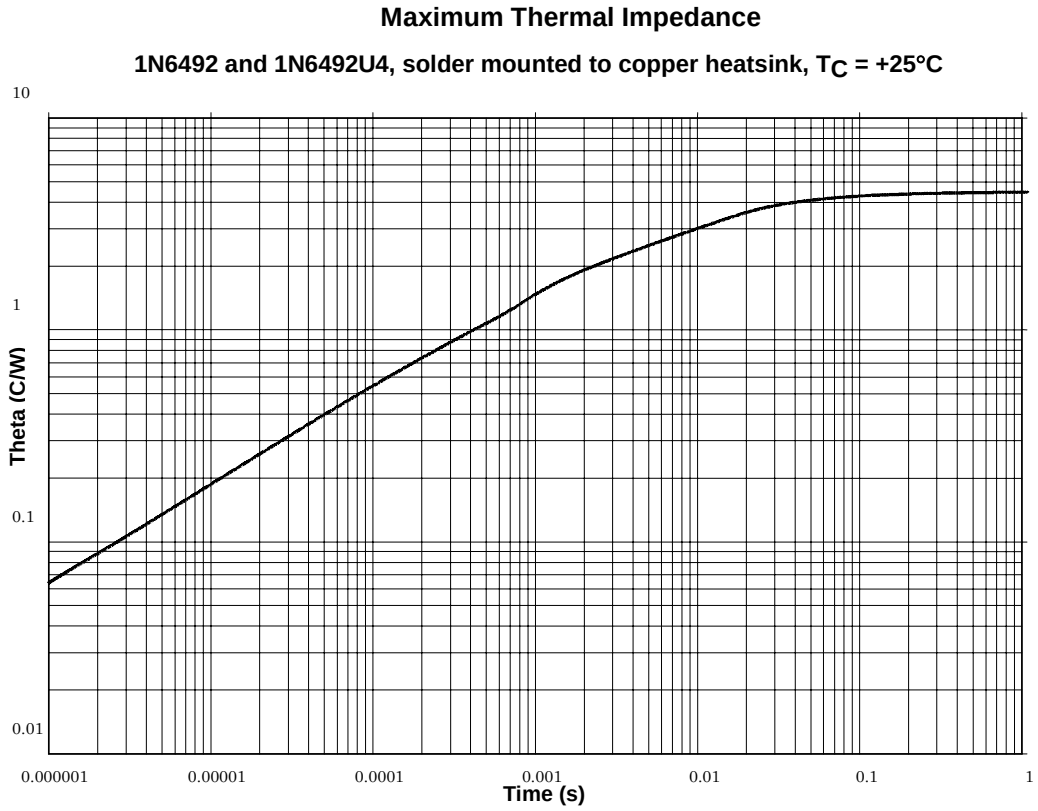
1/ For sampling plan see MIL-PRF-19500.
2/ Measurement point is on the lead .2 inch (5.08 mm) below the seating plane of the case.

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* TABLE II. Groups B and C delta tests. 1/ 2/

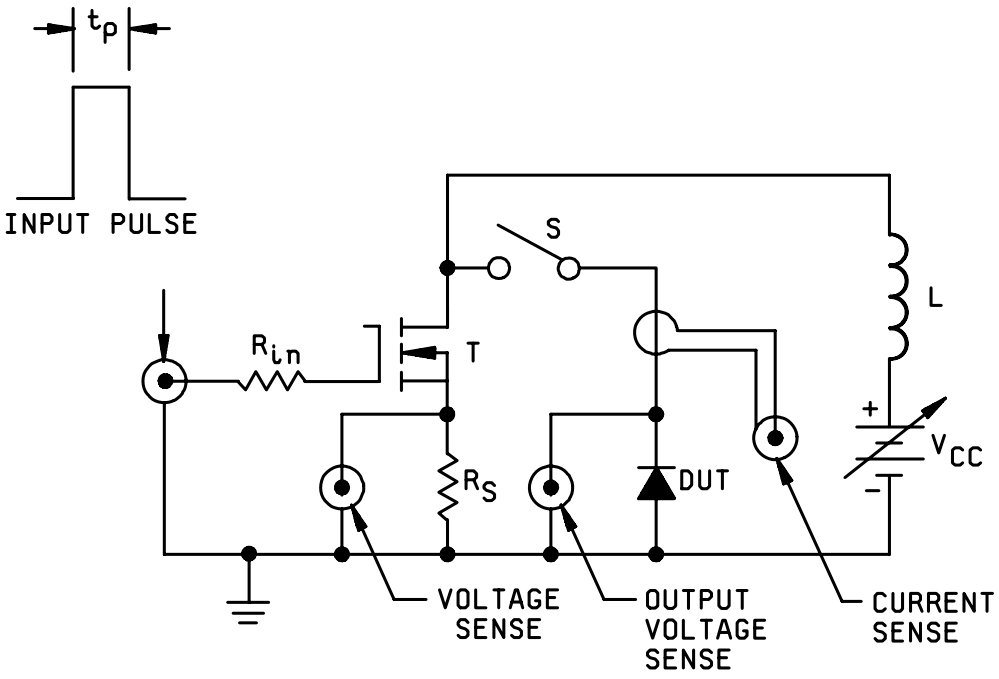
Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_{FM} = 4 A(pk) pulsed (see 4.5.1)	ΔV_{FM2}	±50 mV change from previous value		V
2	Reverse current	4016	V_{RM} = 45 V(pk) pulsed (see 4.5.1)	ΔI_{RM1}	+500 μA dc or +100 percent of initial value, whichever is greater		mV

1/ The delta measurements for table E-VIa (JANS) of MIL-PRF-19500 are as follows: Subgroups 4 and 5, see table II herein, steps 1 and 2.
2/ The delta measurements for table E-VII of MIL-PRF-19500 are as follows: Subgroup 6, see table II herein, steps 1 and 2.



* FIGURE 3. Thermal impedance graph ($R_{\Theta JC}$) for 1N6492 and 1N6492U4 (TO-205AF and U4).

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V_G = 10 Volts	R_{in} = 5.0 Ω, 1 W
R_G = 50 Ω	R_S = 0-1 Ω, 1 W
PW = 30 μs	L = 260 μH
Duty cycle ≤ 1 percent	T = IRF130/2N6756 or equivalent

Procedure:

1. With S open, adjust pulse width to test current of 2 A across R_S.
2. Close S. Verify test current with current sense.
3. Read peak output voltage (see 4.5.4).

FIGURE 4. Peak reverse energy test circuit.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

* 6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Lead finish (see 3.4.1).

c. Product assurance level and type designator.

d. Packaging requirements (see 5.1).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil .

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI
Navy - MC, SH
Air Force - 19, 71, 99

Preparing activity:
DLA - CC

(Project 5961-2007-012)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .

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